

Daily Problem Solutions Booklet

Comprehensive Step-by-Step Analytical Solutions

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Sunday 22 March 2026

1.1 Daily Limit

Limit Involving an Integral of Logarithm

Evaluate the limit:

$$\lim_{x \rightarrow 0} \frac{\int_0^x \ln(1+t^2) dt}{x^3}$$

Solution: Let L denote the limit. As $x \rightarrow 0$, both the numerator and the denominator approach 0, yielding an indeterminate form of type $\frac{0}{0}$. We apply L'Hôpital's Rule. By the First Fundamental Theorem of Calculus, the derivative of the numerator with respect to x is:

$$\frac{d}{dx} \int_0^x \ln(1+t^2) dt = \ln(1+x^2)$$

The derivative of the denominator is $\frac{d}{dx}(x^3) = 3x^2$. Applying L'Hôpital's Rule gives:

$$L = \lim_{x \rightarrow 0} \frac{\ln(1+x^2)}{3x^2}$$

We use the standard Maclaurin series expansion for the natural logarithm, $\ln(1+u) = u - \frac{u^2}{2} + O(u^3)$ as $u \rightarrow 0$. Substituting $u = x^2$, we obtain:

$$\ln(1+x^2) = x^2 + O(x^4)$$

Substituting this expansion back into the limit:

$$L = \lim_{x \rightarrow 0} \frac{x^2 + O(x^4)}{3x^2} = \lim_{x \rightarrow 0} \left(\frac{1}{3} + O(x^2) \right) = \frac{1}{3}$$

Monday 23 March 2026

2.1 Daily Limit

Limit of Gamma Function Ratio with Exponential Scaling

Evaluate the limit:

$$\lim_{x \rightarrow \infty} \left(\frac{\Gamma\left(x + \frac{1}{x}\right)}{\Gamma\left(x - \frac{1}{x}\right)} \right)^{\frac{x}{\ln(x)}}$$

Solution: Let L denote the required limit. We first analyze the asymptotic behavior of the ratio of the Gamma functions as $x \rightarrow \infty$. A well-known asymptotic consequence of Stirling's approximation is that for constants (or sufficiently small perturbations) a and b :

$$\frac{\Gamma(x+a)}{\Gamma(x+b)} \sim x^{a-b} \quad \text{as } x \rightarrow \infty$$

Here, $a = \frac{1}{x}$ and $b = -\frac{1}{x}$. Thus, the difference is $a - b = \frac{2}{x}$. Using a more precise logarithmic expansion with the Digamma function $\psi(x) \sim \ln x$:

$$\ln \Gamma\left(x + \frac{1}{x}\right) - \ln \Gamma\left(x - \frac{1}{x}\right) \approx \frac{2}{x} \psi(x) \approx \frac{2 \ln x}{x}$$

Taking the exponential yields the equivalence for the ratio:

$$\frac{\Gamma\left(x + \frac{1}{x}\right)}{\Gamma\left(x - \frac{1}{x}\right)} \sim x^{\frac{2}{x}} = e^{\frac{2 \ln x}{x}}$$

Now, we substitute this back into the original limit expression:

$$L = \lim_{x \rightarrow \infty} \left(e^{\frac{2 \ln x}{x}} \right)^{\frac{x}{\ln x}}$$

Using the properties of exponents, we multiply the powers:

$$L = \lim_{x \rightarrow \infty} e^{\frac{2 \ln x}{x} \cdot \frac{x}{\ln x}} = e^2$$

Tuesday 24 March 2026

3.1 Daily Limit

Limit of Exponential and Trigonometric Difference

Evaluate the limit:

$$\lim_{x \rightarrow 0} \frac{e^{-2x} - \cos(x^2)}{x \cos(x)}$$

Solution: We utilize the Taylor series expansions of the functions involved around $x = 0$:

1. For the exponential function:

$$e^{-2x} = 1 - 2x + \frac{(-2x)^2}{2!} - \dots = 1 - 2x + 2x^2 + O(x^3)$$

2. For the cosine term in the numerator:

$$\cos(x^2) = 1 - \frac{(x^2)^2}{2!} + \dots = 1 - \frac{x^4}{2} + O(x^8)$$

3. For the denominator:

$$x \cos(x) = x \left(1 - \frac{x^2}{2} + O(x^4) \right) = x - \frac{x^3}{2} + O(x^5)$$

Now, we substitute these expansions into the numerator:

$$e^{-2x} - \cos(x^2) = (1 - 2x + 2x^2 + O(x^3)) - \left(1 - \frac{x^4}{2} + O(x^8) \right) = -2x + 2x^2 + O(x^3)$$

Next, divide by the Taylor expansion of the denominator:

$$L = \lim_{x \rightarrow 0} \frac{-2x + 2x^2 + O(x^3)}{x - \frac{x^3}{2} + O(x^5)}$$

Factoring out x from both the numerator and the denominator (valid since $x \rightarrow 0$ implies $x \neq 0$):

$$L = \lim_{x \rightarrow 0} \frac{-2 + 2x + O(x^2)}{1 - \frac{x^2}{2} + O(x^4)} = \frac{-2}{1} = -2$$

Wednesday 25 March 2026

4.1 Daily Limit

Double Sum Limit to Double Integral

Evaluate the limit:

$$\lim_{n \rightarrow \infty} \sum_{j,k \geq 1}^n \frac{1}{n^2 + n(j+k)}$$

Solution: We rewrite the summation to expose a 2D Riemann sum structure. Factoring out $\frac{1}{n^2}$ from the general term yields:

$$\sum_{j=1}^n \sum_{k=1}^n \frac{1}{n^2 \left(1 + \frac{j}{n} + \frac{k}{n}\right)} = \frac{1}{n^2} \sum_{j=1}^n \sum_{k=1}^n \frac{1}{1 + \left(\frac{j}{n}\right) + \left(\frac{k}{n}\right)}$$

As $n \rightarrow \infty$, this double sum converges to the double integral over the unit square $[0, 1] \times [0, 1]$:

$$I = \int_0^1 \int_0^1 \frac{1}{1+x+y} dx dy$$

We first evaluate the inner integral with respect to x :

$$\int_0^1 \frac{1}{1+y+x} dx = \left[\ln(1+y+x) \right]_{x=0}^{x=1} = \ln(2+y) - \ln(1+y)$$

Next, we integrate this result with respect to y . We use the standard integration formula $\int \ln(u) du = u \ln(u) - u$:

$$I = \int_0^1 \ln(2+y) dy - \int_0^1 \ln(1+y) dy$$

$$I = \left[(2+y) \ln(2+y) - (2+y) \right]_0^1 - \left[(1+y) \ln(1+y) - (1+y) \right]_0^1$$

Evaluating the first bracket:

$$(3 \ln 3 - 3) - (2 \ln 2 - 2) = 3 \ln 3 - 2 \ln 2 - 1$$

Evaluating the second bracket:

$$(2 \ln 2 - 2) - (1 \ln 1 - 1) = 2 \ln 2 - 1$$

Subtracting the two results gives the final value:

$$I = (3 \ln 3 - 2 \ln 2 - 1) - (2 \ln 2 - 1) = 3 \ln 3 - 4 \ln 2 = \ln \left(\frac{27}{16} \right)$$

Thursday 26 March 2026

5.1 Daily Limit

Limit of Difference Involving Euler's Number

Evaluate the limit:

$$\lim_{x \rightarrow \infty} x \left(\left(1 + \frac{1}{x} \right)^x - e \right)$$

Solution: We must find the precise asymptotic expansion of $\left(1 + \frac{1}{x} \right)^x$ as $x \rightarrow \infty$. We rewrite it using the exponential and natural logarithm functions:

$$\left(1 + \frac{1}{x} \right)^x = \exp \left(x \ln \left(1 + \frac{1}{x} \right) \right)$$

We substitute the Taylor series for $\ln(1+u) = u - \frac{u^2}{2} + \frac{u^3}{3} - \dots$ with $u = \frac{1}{x}$:

$$x \ln \left(1 + \frac{1}{x} \right) = x \left(\frac{1}{x} - \frac{1}{2x^2} + O \left(\frac{1}{x^3} \right) \right) = 1 - \frac{1}{2x} + O \left(\frac{1}{x^2} \right)$$

Exponentiating this expansion gives:

$$\exp \left(1 - \frac{1}{2x} + O \left(\frac{1}{x^2} \right) \right) = e \cdot e^{-\frac{1}{2x} + O \left(\frac{1}{x^2} \right)}$$

Applying the Taylor series $e^v = 1 + v + O(v^2)$ with $v = -\frac{1}{2x} + O \left(\frac{1}{x^2} \right)$:

$$e \left(1 - \frac{1}{2x} + O \left(\frac{1}{x^2} \right) \right) = e - \frac{e}{2x} + O \left(\frac{1}{x^2} \right)$$

Now substitute this back into the limit expression:

$$L = \lim_{x \rightarrow \infty} x \left(\left(e - \frac{e}{2x} + O \left(\frac{1}{x^2} \right) \right) - e \right)$$

The e terms cancel out, leaving:

$$L = \lim_{x \rightarrow \infty} x \left(-\frac{e}{2x} + O \left(\frac{1}{x^2} \right) \right) = \lim_{x \rightarrow \infty} \left(-\frac{e}{2} + O \left(\frac{1}{x} \right) \right) = -\frac{e}{2}$$

Friday 27 March 2026

6.1 Daily Limit

Limit with Unknown Parameters and Square Root

If $\lim_{x \rightarrow b} \frac{4 - \sqrt{a(x+b)}}{b-x} = b$, where $a < 0, b < 0$, find the value of $a - b$.

Solution: For the limit to exist and be finite while the denominator $b - x \rightarrow 0$ as $x \rightarrow b$, the numerator must also approach 0. Setting the limit of the numerator to 0 gives:

$$4 - \sqrt{a(b+b)} = 0 \implies 4 = \sqrt{2ab} \implies 16 = 2ab \implies ab = 8$$

This implies a and b must have the same sign; the condition $a < 0$ and $b < 0$ is consistent with this. Because we have a $\frac{0}{0}$ indeterminate form, we can apply L'Hôpital's Rule:

$$\lim_{x \rightarrow b} \frac{\frac{d}{dx} (4 - \sqrt{a(x+b)})}{\frac{d}{dx} (b-x)} = \lim_{x \rightarrow b} \frac{-\frac{a}{2\sqrt{a(x+b)}}}{-1} = \frac{a}{2\sqrt{2ab}}$$

Since we are given that the limit evaluates to b , we have:

$$\frac{a}{2\sqrt{2ab}} = b$$

From our earlier result, we know $ab = 8$, so $\sqrt{2ab} = \sqrt{16} = 4$. Substituting this in:

$$\frac{a}{2(4)} = b \implies \frac{a}{8} = b \implies a = 8b$$

Substitute $a = 8b$ into the relation $ab = 8$:

$$(8b)(b) = 8 \implies 8b^2 = 8 \implies b^2 = 1$$

Given $b < 0$, we take the negative root: $b = -1$. Consequently, $a = 8(-1) = -8$. (Note that $a = -8 < 0$, matching the given conditions). Finally, we calculate the requested quantity $a - b$:

$$a - b = -8 - (-1) = -7$$

Saturday 28 March 2026

7.1 Daily Limit

Limit Involving Natural Logarithm of Partial Sum of e

Evaluate the limit:

$$\lim_{k \rightarrow \infty} \left(1 - \ln \sum_{n=0}^k \frac{1}{n!} \right) \cdot k!$$

Solution: Let $S_k = \sum_{n=0}^k \frac{1}{n!}$ be the k -th partial sum of the Taylor series for e . The remainder term is given by:

$$e - S_k = \sum_{n=k+1}^{\infty} \frac{1}{n!} = \frac{1}{(k+1)!} + \frac{1}{(k+2)!} + \dots$$

For large k , we can approximate this by factoring out the first term:

$$e - S_k = \frac{1}{(k+1)!} \left(1 + \frac{1}{k+2} + \dots \right) = \frac{1}{(k+1)!} + O\left(\frac{1}{(k+2)!}\right)$$

So we can write $S_k = e - \frac{1}{(k+1)!} + O\left(\frac{1}{(k+2)!}\right)$. Now evaluate the natural logarithm of this partial sum:

$$\ln(S_k) = \ln\left(e \left(1 - \frac{1}{e(k+1)!} + \dots\right)\right) = 1 + \ln\left(1 - \frac{1}{e(k+1)!} + \dots\right)$$

Using the Taylor expansion $\ln(1 - u) \sim -u$ for small u :

$$\ln(S_k) = 1 - \frac{1}{e(k+1)!} + O\left(\frac{1}{(k+2)!}\right)$$

Substitute this back into the parenthetical term:

$$1 - \ln(S_k) = 1 - \left(1 - \frac{1}{e(k+1)!} + \dots\right) = \frac{1}{e(k+1)!} + O\left(\frac{1}{(k+2)!}\right)$$

Multiplying this expression by $k!$:

$$\left(1 - \ln \sum_{n=0}^k \frac{1}{n!}\right) \cdot k! = k! \left(\frac{1}{e(k+1)!} + \dots\right) = \frac{k!}{e(k+1)k!} + \dots = \frac{1}{e(k+1)} + O\left(\frac{1}{(k+1)^2}\right)$$

Taking the limit as $k \rightarrow \infty$:

$$L = \lim_{k \rightarrow \infty} \frac{1}{e(k+1)} = 0$$

Sunday 29 March 2026

8.1 Daily Limit

Composite Function Limit with Logarithmic Behavior

Evaluate $\lim_{x \rightarrow \alpha^+} f(g(x))$, where

$$g(x) = \frac{2 \log_e(\sqrt{x} - \sqrt{\alpha})}{\log_e(e^{\sqrt{x}} - e^{\sqrt{\alpha}})} \quad \text{and} \quad f(y) = \sin\left(\frac{\pi y}{12}\right)$$

Solution: Since $f(y)$ is continuous, $\lim_{x \rightarrow \alpha^+} f(g(x)) = f(\lim_{x \rightarrow \alpha^+} g(x))$. We first evaluate the limit of the inner function $g(x)$. Let $t = \sqrt{x} - \sqrt{\alpha}$. As $x \rightarrow \alpha^+$, we have $t \rightarrow 0^+$. Consequently, $\sqrt{x} = t + \sqrt{\alpha}$. Substituting t into $g(x)$, we rewrite the denominator:

$$\ln(e^{t+\sqrt{\alpha}} - e^{\sqrt{\alpha}}) = \ln(e^{\sqrt{\alpha}}(e^t - 1)) = \sqrt{\alpha} + \ln(e^t - 1)$$

The numerator simply becomes $2 \ln(t)$. Thus:

$$g(x) = \frac{2 \ln(t)}{\sqrt{\alpha} + \ln(e^t - 1)}$$

For t very close to 0, the Maclaurin expansion $e^t - 1 \sim t$ holds. Therefore, $\ln(e^t - 1) \sim \ln(t)$:

$$\lim_{t \rightarrow 0^+} g(x) = \lim_{t \rightarrow 0^+} \frac{2 \ln(t)}{\sqrt{\alpha} + \ln(t)}$$

Divide both the numerator and the denominator by $\ln(t)$:

$$\lim_{t \rightarrow 0^+} g(x) = \lim_{t \rightarrow 0^+} \frac{2}{\frac{\sqrt{\alpha}}{\ln(t)} + 1}$$

As $t \rightarrow 0^+$, $\ln(t) \rightarrow -\infty$, meaning $\frac{\sqrt{\alpha}}{\ln(t)} \rightarrow 0$. Therefore:

$$\lim_{x \rightarrow \alpha^+} g(x) = \frac{2}{0 + 1} = 2$$

Now apply the outer continuous function $f(y)$:

$$L = f(2) = \sin\left(\frac{2\pi}{12}\right) = \sin\left(\frac{\pi}{6}\right) = \frac{1}{2}$$

Monday 30 March 2026

9.1 Daily Limit

Limit of a Sum via Riemann Integral

Evaluate the limit:

$$\lim_{n \rightarrow \infty} \left(\frac{n^2}{n^4 + 1} + \frac{2n^2}{n^4 + 16} + \cdots + \frac{1}{2n} \right)$$

Solution: We begin by expressing the series in summation notation. The k -th term of the series takes the form $\frac{k \cdot n^2}{n^4 + k^4}$. Notice that for the final term, when $k = n$, the term is $\frac{n \cdot n^2}{n^4 + n^4} = \frac{n^3}{2n^4} = \frac{1}{2n}$, which matches the problem description exactly.

$$S_n = \sum_{k=1}^n \frac{kn^2}{n^4 + k^4}$$

To turn this into a Riemann sum, we factor out $\frac{1}{n}$ from the general term and divide the numerator and denominator by n^4 :

$$S_n = \sum_{k=1}^n \frac{1}{n} \cdot \frac{kn^3}{n^4 + k^4} = \frac{1}{n} \sum_{k=1}^n \frac{\left(\frac{k}{n}\right)}{1 + \left(\frac{k}{n}\right)^4}$$

Let $x_k = \frac{k}{n}$ with $\Delta x = \frac{1}{n}$. As $n \rightarrow \infty$, the Riemann sum converges to the definite integral over $[0, 1]$:

$$L = \int_0^1 \frac{x}{1 + x^4} dx$$

To evaluate this integral, we employ the substitution $u = x^2$, making $du = 2x dx$, or $\frac{1}{2}du = x dx$. The limits of integration remain 0 to 1:

$$L = \frac{1}{2} \int_0^1 \frac{1}{1 + u^2} du = \frac{1}{2} \left[\arctan(u) \right]_0^1$$

Evaluating the boundaries:

$$L = \frac{1}{2} (\arctan(1) - \arctan(0)) = \frac{1}{2} \left(\frac{\pi}{4} - 0 \right) = \frac{\pi}{8}$$

Tuesday 31 March 2026

10.1 Daily Limit

Nested Inverse Tangent and Sine Limit

Evaluate the limit:

$$\lim_{x \rightarrow 0} \frac{\tan^{-1}(\tan^{-1}(\tan^{-1}(x)))}{\sin(\sin(\sin(x)))}$$

Solution: To find this limit without repeatedly applying L'Hôpital's Rule on complex nested chains, we utilize the first-order asymptotic equivalences of these functions near zero. It is a well-known fact that as $u \rightarrow 0$:

$$\tan^{-1}(u) \sim u \quad \text{and} \quad \sin(u) \sim u$$

We establish the behavior of the nested functions. Let $f_1(x) = \tan^{-1}(x) \sim x$. Because $f_1(x) \rightarrow 0$ as $x \rightarrow 0$, we can substitute it into the equivalence:

$$f_2(x) = \tan^{-1}(f_1(x)) \sim f_1(x) \sim x$$

Applying it a third time for the numerator gives:

$$\tan^{-1}(\tan^{-1}(\tan^{-1}(x))) \sim x$$

By identical recursive logic for the denominator, given that $\sin(x) \sim x$ as $x \rightarrow 0$:

$$\sin(\sin(\sin(x))) \sim x$$

Using these leading-order terms directly in the limit quotient:

$$L = \lim_{x \rightarrow 0} \frac{\tan^{-1}(\tan^{-1}(\tan^{-1}(x)))}{\sin(\sin(\sin(x)))} = \lim_{x \rightarrow 0} \frac{x}{x} = 1$$